

# US Patent & Trademark Office

## Patent Public Search | Text View

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United States Patent Application Publication

20250276564

Kind Code

A1

Publication Date

September 04, 2025

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### WINDOW ASSEMBLIES HAVING TWO OR MORE OPEN CONFIGURATIONS

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#### Abstract

An example window assembly includes a fixed pane, a first movable pane movable between a closed position and an open position, and a second movable pane movable between a closed position and a first partially open position where at least a portion of the second movable pane is disposed over the fixed pane, between the closed position and a second partially open position where at least a portion of the second movable pane is disposed over the first movable pane, and between the closed position and the open position.

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**Appl. No.:** 18/592708

**Filed:** March 01, 2024

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#### Publication Classification

**Int. Cl.:** B60J1/18 (20060101); B60J1/00 (20060101); E05C1/00 (20060101); E05C1/16 (20060101)

**U.S. Cl.:**

**CPC** B60J1/1861 (20130101); B60J1/007 (20130101); E05C1/004 (20130101); E05C1/16 (20130101);

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## Background/Summary

### FIELD OF THE DISCLOSURE

[0001] Example embodiments relate to window assemblies (for example, for use in vehicles, including construction vehicles, like backhoe loaders) having two or more open configurations.

### BACKGROUND

[0002] Window systems for construction or utility vehicles (such as backhoe loaders) often have an all-or-nothing approach, where the window is either fully closed or fully opened.

### SUMMARY

[0003] In various aspects, the present disclosure provides an example window assembly having two or more open configurations.

[0004] In at least one example embodiment, the window assembly includes a fixed pane, a movable pane, and a third pane, where the movable pane is at least partially movable over the fixed pane and at least partially movable over the third pane.

[0005] In at least one example embodiment, the window assembly may further include one or more latch assemblies coupled to the movable pane.

[0006] In at least one example embodiment, each of the one or more latch assemblies may include a tongue member that is movable between an extended position and a retracted position.

[0007] In at least one example embodiment, the window assembly may further include a track assembly configured to support the movable pane, where the track assembly includes a supporting portion and a locking portion. The supporting portion may be configured to receive at least a portion of the movable pane, and the locking portion may include two or more notches configured to receive the tongue member of a respective latch assembly of the one or more latch assemblies.

[0008] In at least one example embodiment, the track assembly may be a first track assembly, and the window assembly may further include a second track assembly configured to receive at least a portion of the fixed pane.

[0009] In at least one example embodiment, the movable pane may be a first movable pane and the third pane may be a second movable pane, the second movable pane may be movable between a first position and an overhead position.

[0010] In at least one example embodiment, the first movable pane may be moveable over the second movable pane and into the overhead position with the second movable pane.

[0011] In at least one example embodiment, the window assembly may further include one or more latch assemblies in association with the second movable pane.

[0012] In at least one example embodiment, each of the one or more latch assemblies may include a tongue member that is movable between an extended position and a retracted position.

[0013] In at least one example embodiment, the window assembly may further include a track assembly configured to support the second movable pane, where the track assembly includes a supporting portion and a locking portion. The supporting portion may be configured to receive at least a portion of the second movable pane, and the locking portion may include one or more notches configured to receive the respective tongue member of the respective latch assembly of the one or more latch assemblies.

[0014] In various aspects, the present disclosure provides another example window assembly having two or more open configurations.

[0015] In at least one example embodiment, the window assembly includes a fixed pane, a first movable pane movable between a closed position and an open position, and a second movable pane movable between a closed position and a first partially open position where at least a portion of the second movable pane is disposed over the fixed pane, between the closed position and a second partially open position where at least a portion of the second movable pane is disposed over the

first movable pane, and between the closed position and the open position.

[0016] In at least one example embodiment, the window assembly may further include one or more latch assemblies associated with the first movable pane.

[0017] In at least one example embodiment, the each of the one or more latch assemblies may include a tongue member that is movable between an extended position and a retracted position.

[0018] In at least one example embodiment, the window assembly may further include a track assembly configured to support the first movable pane, where the track assembly itself may be configured to move between the closed position of the first movable pane to the open position of the first movable pane.

[0019] In at least one example embodiment, the window assembly may further include one or more latch assemblies coupled to the second movable pane.

[0020] In at least one example embodiment, the each of the one or more latch assemblies may include a tongue member that is movable between an extended position and a retracted position.

[0021] In at least one example embodiment, the window assembly may further includes a track assembly configured to support the second movable pane, where the track assembly includes a supporting portion and a locking portion. The supporting portion may be configured to receive at least a portion of the second movable pane, and the locking portion may include one or more first notches corresponding with the closed position, one or more second notches corresponding with the first partially open position, one or more third notches corresponding with the second partially open position, and one or more fourth notches corresponding with the open position, where each notch of the one or more first notches, of the one or more second notches, of the one or more third notches, and of the one or more fourth notches is configured to receive the respective tongue member of the one or more latch assemblies.

[0022] In various aspects, the present disclosure provides another example window assembly having two or more open configurations.

[0023] In at least one example embodiment, the window assembly may include a fixed pane, a first pane assembly, and a second pane assembly. The first pane assembly may include a first movable pane that is movable between a closed position and an overhead position, a first latch assembly associated with the first movable pane, and a first track assembly configured itself to move between the closed position of the first movable pane to the overhead position of the first movable pane. The second pane assembly may include a second movable pane that is movable between a closed position and a first partially open position where at least a part of the second movable pane is disposed over a portion of the fixed pane, between the closed position and a second partially open position where at least a part of the second movable pane is disposed over a portion of the fixed movable pane, and between the closed position and the overhead position. The second pane assembly may also include a second latch assembly coupled to the second movable pane and a second track assembly. The second track assembly may include a supporting portion configured to receive at least a portion of the second movable pane and a locking portion including four or more notches configured to receive a portion of the second latch assembly.

[0024] In at least one example embodiment, a strut may enable the movement of the first track assembly between the closed position of the first movable pane and the overhead position of the first movable pane.

[0025] In at least one example embodiment, the second latch assembly may include a tongue member that is movable between an extended position and a retracted position, and the four or more notches may be configured to receive at least a portion of the tongue member.

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## Description

## BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The various features and advantages of the non-limiting embodiments herein may become more apparent upon review of the detailed description in conjunction with the accompanying drawings. The accompanying drawings are merely provided for illustrative purposes and should not be interpreted to limit the scope of the claims. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. For the purposes of clarity, various dimensions of the drawings may have been exaggerated.

[0027] FIG. 1 is a perspective view of an example vehicle having a window assembly that includes three panes (where two of the three panes are moveable a track assembly and the remaining pane of the three panes is fixed) and four latch assemblies in accordance with at least one example embodiment of the present disclosure.

[0028] FIG. 2 is an interior view of the window assembly as illustrated in FIG. 1, where the window assembly is in a fully closed state in accordance with at least one example embodiment of the present disclosure.

[0029] FIG. 3A is a perspective view of an example latch assembly that may be used as one or more of the latch assemblies of the window assembly as illustrated in FIG. 2 in accordance with at least one example embodiment of the present disclosure.

[0030] FIG. 3B is a partial exploded view of the example latch assembly illustrated in FIG. 3A in accordance with at least one example embodiment of the present disclosure.

[0031] FIG. 4 is a perspective view of the track assembly of the window assembly as illustrated in FIG. 2, where the track assembly includes a first and second movable tracks and a third fixed track, the first and second movable tracks including a plurality of notches configured to interact with the latch assemblies in accordance with at least one example embodiment of the present disclosure.

[0032] FIG. 4A is a perspective view of a first portion of the track assembly of FIG. 4 in accordance with at least one example embodiment of the present disclosure.

[0033] FIG. 4B is a perspective view of a second portion of the track assembly of FIG. 4 in accordance with at least one example embodiment of the present disclosure.

[0034] FIG. 4C is a perspective view of a third portion of the track assembly of FIG. 4 in accordance with at least one example embodiment of the present disclosure.

[0035] FIG. 5A is an interior view of the window assembly as illustrated in FIG. 2, where the movable, middle pane and the movable, top pane are stored in an overhead position, and the window is in a fully open state in accordance with at least one example embodiment of the present disclosure.

[0036] FIG. 5B is a perspective view of the window assembly as illustrated in FIG. 5A, where the movable, middle pane and the movable, top pane are stored in an overhead position, and the window is in a fully open state in accordance with at least one example embodiment of the present disclosure.

[0037] FIG. 6 is an interior view of the window assembly as illustrated in FIG. 2, where at least part of the movable, middle pane is stored over at least part of a fixed, bottom pane, and the window is in a (first) partially open state in accordance with at least one example embodiment of the present disclosure.

[0038] FIG. 7 is a cross-sectional view of the window assembly, and more specifically, the second track assembly supporting the movable, middle pane as illustrated in FIG. 6, in accordance with at least one example embodiment of the present disclosure.

[0039] FIG. 8 is an interior view of the window assembly as illustrated in FIG. 2, where at least part of a movable, middle pane of the three panes is stored over a movable, top pane of the three panes, and the window is in a (second) partially open state in accordance with at least one example embodiment of the present disclosure.

[0040] FIG. 9 is a cross-sectional view of the window assembly, and more specifically, the second

track assembly supporting the movable, middle pane as illustrated in FIG. 8, in accordance with at least one example embodiment of the present disclosure.

[0041] FIG. 10A is an interior view of the window assembly as illustrated in FIG. 2, where at least part of the movable, top pane is stored overhead, and the window is in a (third) partially open state in accordance with at least one example embodiment of the present disclosure.

[0042] FIG. 10B is a perspective view of the window assembly as illustrated in FIG. 10A, where at least part of the movable, top pane is stored overhead, and the window is in a (third) partially open state in accordance with at least one example embodiment of the present disclosure.

[0043] FIG. 11 is a view of the window assembly as illustrated in FIG. 1, where at least part of the movable, top pane is stored overhead and at least part of the movable, middle pane is stored over a fixed, bottom pane, and the window is in a (fourth) partially open state in accordance with at least one example embodiment of the present disclosure.

#### DETAILED DESCRIPTION

[0044] Some detailed example embodiments are disclosed herein. However, specific structural and functional details disclosed herein are merely representative for the purposes of describing example embodiments. Example embodiments may, however, be embodied in many alternate forms and should not be construed as limited to only the example embodiments set forth herein.

[0045] Accordingly, while example embodiments are capable of various modifications and alternative forms, example embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit example embodiments to the particular forms disclosed, but to the contrary, example embodiments are to cover all modifications, equivalents, and alternatives falling within the scope of example embodiments. Like numbers refer to like elements throughout the description of the figures.

[0046] It should be understood that when an element or layer is referred to as being “on,” “connected to,” “coupled to,” or “covering” another element or layer, it may be directly on, connected to, coupled to, or covering the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout the specification. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0047] It should be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, regions, layers and/or sections, these elements, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, region, layer, or section from another region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, region, layer, or section without departing from the teachings of example embodiments.

[0048] Spatially relative terms (e.g., “beneath,” “below,” “lower,” “above,” “upper,” and the like) may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It should be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0049] The terminology used herein is for the purpose of describing various example embodiments only and is not intended to be limiting of example embodiments. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,”

“comprises,” and/or “comprising,” specify the presence of stated features, integers, steps, operations, and/or elements, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, and/or groups thereof.

[0050] When the terms “about” or “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value includes a manufacturing or operational tolerance (e.g., +10%) around the stated numerical value. Moreover, when the terms “generally” or “substantially” are used in connection with geometric shapes, it is intended that precision of the geometric shape is not required but that latitude for the shape is within the scope of the disclosure. Furthermore, regardless of whether numerical values or shapes are modified as “about,” “generally,” or “substantially,” it will be understood that these values and shapes should be construed as including a manufacturing or operational tolerance (e.g.,  $\pm 10\%$ ) around the stated numerical values or shapes.

[0051] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, including those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0052] At least some example embodiments of the present disclosure provide window assemblies (for example, for use in vehicles, including construction vehicles, like backhoe loaders) having two or more open configurations.

[0053] As used herein, vehicle may include, for example, an automotive vehicle and/or an agricultural vehicle and/or a mining vehicle and/or a construction vehicle and/or a utility vehicle. The vehicle may include, for example, backhoe loaders, tractors, planters, sprayers, bulldozers, forklifts, steam rollers, cranes, haul truck, underground graders, rock breakers, and the like.

[0054] Window systems for construction or utility vehicles (such as backhoe loaders) often have an all-or-nothing approach, where the window is either fully closed or fully opened, offering the user limited options during inclement weather. The present disclosure provides window assemblies having one or more venting positions.

[0055] FIG. 1 is a perspective view of an example vehicle **100** including a window assembly **110** in accordance with at least one example embodiment of the present disclosure. The window assembly **110** includes a track (or rail) assembly **120** configured to support two movable panes (or panels) **112**, **114** and a single fixed pane (or panel) **116**. The panes **112**, **114**, **116** may be transparent, translucent, or opaque.

[0056] Although the vehicle **100** is illustrated as including only a single window assembly **110**, it should be appreciated that, in other example embodiments, vehicles may include two or more window assemblies. Further, although three panes are illustrated, it should be appreciated that, in other example embodiments, window assemblies may include only two panes where the two panes are moveable over a stationary structure (or component), or in yet other example embodiments, four or more panes where at least two of the panes are movable.

[0057] FIG. 2 is a view of the window assembly **110** in a fully closed state in accordance with at least one example embodiment of the present disclosure. As illustrated, the window assembly **110** has three panes, including a top (or first) pane **112**, a middle (or second) pane **114**, and a bottom (or third) pane **116**, where in a closed position (as illustrated in FIGS. 1 and 2), the middle pane **114** is disposed between the top pane **112** and the bottom pane **116**. The top pane **112** and the middle pane **114** are movable panes, while the bottom pane **116** is a fixed pane. For example, the top pane **112** may move (for example, along a first track **130** of the track assembly **120** as illustrated in FIG. 4) from a first (or closed) position (as illustrated in FIGS. 1 and 2) to second (or open or overhead) position (as illustrated in FIGS. 5 and 10), as further discussed below. The middle pane **114** may move (for example, along a second track **140** of the track assembly **120** as illustrated in FIG. 4)

from a closed position (as illustrated in FIGS. **1** and **2**) to a first open position where the middle pane **114** is stored overhead with the top pane **112** (as illustrated in FIGS. **5A** and **5B**) and/or to a second open position where at least a portion of the middle pane **114** is stored over (in front of or behind) at least a portion of the top pane **112** (as illustrated in FIG. **8**) and/or to a third open position where at least a portion of the middle pane **114** is stored over (in front of or behind) at least a portion of the bottom pane **116** (as illustrated in FIGS. **6** and **11**), each of which being further discussed below. Although the top and second panes **112**, **114** are described as movable panes and the bottom pane **116** is described as a fixed pane, it should be appreciated that, in various other embodiments, the bottom and middle panes may be movable panes while the top pane is a fixed pane.

[0058] With renewed reference to FIG. **2**, the window assembly **110** includes two or more latching (or locking) assemblies **160A**, **160B**, **162A**, **162B**. For example, as illustrated, in at least one example embodiment, the window assembly **110** includes four latching assemblies **160A**, **160B**, **162A**, **162B** (or two sets of latching assemblies), where two of the four latching assemblies are (or a first pair of the two sets of latching assemblies is) associated with the movable, top pane **112** and two of the four latching assemblies are (or a second pair of the latching assemblies is) associated with the movable, middle pane **114**. Although the top pane **112** is illustrated as including two latching assemblies (or a pair of latch assemblies) **160A**, **160B** it should be appreciated that, in various other embodiments, the top pane **112** may be associated with a single latch assembly, or alternatively still, three or more latch assemblies. Although the middle pane **114** is illustrated as including two latching assemblies (or a pair of latch assemblies) **160A**, **160B**, it should be appreciated that, in various other embodiments, the middle pane **114** may be associated with a single latch assembly, or alternatively still, three or more latch assemblies. In at least one example embodiment, each of the latch assemblies **160A**, **160B**, **162A**, **162B** may be the same type having the same configuration.

[0059] The latch assemblies **160A**, **160B** may be disposed along a perimeter of the top pane **112**. For example, in at least one example embodiment, as illustrated, the first and second latch assemblies **160A**, **160B** of the top pane **112** may be associated with opposing sides of the top pane **112**. In at least one example embodiment, as illustrated, the top pane **112** may have a general rectangular shape and the latch assemblies **160A**, **160B** may be associated with the opposing shorter (or smaller) lengths of the rectangular shape. In at least one example embodiment, as illustrated, the latch assemblies **160A**, **160B** may be disposed nearer to a middle point of the respective side than to the starting (or first) end and an ending (or second) end of the respective side.

[0060] The latch assemblies **162A**, **162B** may be disposed along a perimeter of the middle pane **114**. For example, in at least one example embodiment, as illustrated, the first and second latch assemblies **162A**, **162B** of the middle pane **114** may be associated with opposing sides of the middle pane **114**. In at least one example embodiment, as illustrated, the middle pane may have a general rectangular shape and the latch assemblies **162A**, **162B** may be associated with the opposing shorter (or smaller) lengths of the rectangular shape. In at least one example embodiment, as illustrated, the latch assemblies **162A**, **162B** may be disposed nearer to a middle point of the respective side than to the starting (or first) end and an ending (or second) end of the respective side.

[0061] Although the top pane **112**, the middle pane **114**, and the bottom pane **116** are illustrated as each having a general rectangular shape, it should be appreciated that, in other example embodiments, the top pane **112** and/or the middle pane **114** and/or the bottom pane **116** may take the form of one or more other shapes.

[0062] FIG. **3A** is a perspective view of an example latch assembly **300** that may be used as one or more of the latch assemblies **160A**, **160B**, **162A**, **162B**, and FIG. **3B** is a partial exploded view of the window assembly **110** including the example latch assembly **300**.

[0063] The latch assembly **300** includes a handle (or gripping member) **302** and a locking mechanism **310**. In at least one example embodiment, the locking mechanism **310** includes, for example, a trigger **312** in communication with a tongue member **314** such that activation of the trigger **312** may cause the tongue member **314** to move between a first (or extended) position and a second (or retracted) position. In at least one example embodiment, the tongue member **314** may be a spring-loaded mechanism (see spring **315**) and may have at least a portion **317** shaped to be received within one or more portions of the track assembly **120**. For example, as illustrated, the tongue member **314** may be shaped to be received within the notches **146** of the second track **140** of the track assembly **120** that are configured to hold the movable, middle pane **114** in selected positions. Although not specifically illustrated in FIGS. **3A** and **3B**, it should be appreciated that the tongue member **314** may be shaped (additionally or alternatively) to be received the notches **134** of the first track **130** of the track assembly **120** that are configured to hold the movable, top pane **112** in selected positions.

[0064] In at least one example embodiment, the latch assembly **300** may further includes a plate or main body **318** having one or more apertures **320** configured to receive one or more coupling members **322**, for example, for joining the latch assembly **300** to the different panes (e.g., the middle pane **114** as illustrated in FIGS. **3A** and **3B**), as illustrated. In at least one example embodiment, as illustrated in FIG. **3B**, the coupling members **322** may each include a first portion and a second portion, where the first portion protrudes (or extends) through the handle **302**, the main body **318**, the middle pane **114** and joins to the second portion on another side of the middle pane **114**.

[0065] Although not specifically illustrated in FIGS. **3A** and **3B**, it should be appreciated that the main body **318** and coupling members **322** may be similarly used to secure the latch assembly **300** to the receiving plate **136** of the first track **130**. Although the latch assembly **300** is illustrated as including two apertures **320**, it should be appreciated that, in various example embodiments, the latch assembly **300** may include a single aperture or three or more apertures. Further still, it should be appreciated that, in various example embodiments, the latch assembly **320** may be an integrally formed member, for example, formed as an integral member with a pane member (e.g., the middle pane **114**) or a receiving plate (e.g., the receiving plate **136** of the first track **130**).

[0066] FIG. **4** is a perspective view of the track assembly **120**, while FIG. **4A** is a perspective view of a first portion of the track assembly **120**, FIG. **4B** is a perspective view of a second portion of the track assembly **120**, and FIG. **4C** is a perspective view of a third portion of the track assembly **120**. As illustrated, the track assembly **120** includes three tracks (or rails), including a first track (or rail) **130** configured to support the top pane **112**, a second track (or rail) **140** configured to support the middle pane **114**, and a third track (or rail) **150** configured to support the bottom pane **116**.

[0067] The first track **130** may extend, for example, along only a top (or first) portion (or end) **124** of the track assembly **120** (or window assembly **110**). The first track **130** may have a linear shape that is continuous or discontinuous.

[0068] In at least one example embodiment, the first track **130** (e.g., a supporting portion **132** of the first track **130**) supports the top pane **112** by receiving at least a portion of the top pane **112**. In at least one example embodiment, the first track **130** supports the top pane **112** by being formed integrally with the top pane **112**. In at least one example embodiment, the first track **130** supports the top pane **112** by being couplable one or more portions of the top pane **112**.

[0069] The second track **140** may extend, for example, along an entire length of the track assembly **120** (or window assembly **110**). The second track **140** may have a linear shape that is continuous or discontinuous. For example, as illustrated, in at least one example embodiment, the second track **140** may be discontinuous so as to allow a first portion (or half) **140A** of the second track **140** to secure the top pane **112** and/or the middle pane **114** in the overhead position (see FIGS. **5A**, **5B**, **10A**, **10B**, and **11**) and the second portion (or half) **140B** of the second track **140** to secure the middle pane **114** in a closed position for the middle pane **114**. In at least one example embodiment,



the first track **130** may be joined to or joinable to the first portion **140A** of the second track **140**.  
[0070] In at least one example embodiment, the second track **140** (e.g., a supporting portion **142** of the second track **140**) supports the middle pane **114** by receiving at least a portion of the middle pane **114**. In at least one example embodiment, the second track **140** supports the middle pane **114** by being formed integrally with the middle pane **114**. In at least one example embodiment, the second track **140** supports the middle pane **114** by being couplable one or more portions of the middle pane **114**.

[0071] The third track **150** may extend, for example, along a bottom (or second) portion (or end) **126** of the track assembly **120** (or window assembly **110**) that opposes the top portion **124**. The third track **150** may have a linear shape that is continuous or discontinuous.

[0072] In at least one example embodiment, the third track **150** supports the bottom pane **116** by receiving at least a portion of the bottom pane **116**. In at least one example embodiment, the third track **150** supports the bottom pane **116** by being formed integrally with the bottom pane **116**. In at least one example embodiment, the third track **150** supports the bottom pane **116** by being couplable one or more portions of the bottom pane **116**.

[0073] In at least one example embodiment, the third track **150** is configured to fixedly secure the bottom pane **116**, while the first and second tracks **130**, **140** are configured to allow for movement of the top and middle panes **112**, **114**, respectively.

[0074] As illustrated, for example, in FIGS. **4** and **4A-4C** (and also FIGS. **7** and **9** as discussed below), the second track **140** may include a supporting portion **142** and a locking portion **144**, where the supporting portion **142** supports of the middle pane **114** and the (second) locking portion **144** includes a (second) plurality of notches **146** configured (e.g., shaped) to receive a tongue member of respective latch assemblies **162A**, **162B**. When the respective tongue member is (or tongue members are) retracted, for example, using a trigger assembly of the respective latch assembly, the corresponding pane (i.e., the middle pane **114**) may be released and made movable.

[0075] In at least one example embodiment, the middle pane **114** may be repositioned, for example, in an overhead position with the top pane **112**, as illustrated in FIGS. **5A** and **5B**. In at least one example embodiment, the middle pane **114** may be repositioned, for example, in a (first) partially open or vent state, as illustrated in FIG. **6**. In at least one example embodiment, the middle pane **114** may be repositioned, for example, in a (second) partially open or vent state, as illustrated in FIG. **8**. In at least one example embodiment, the middle pane **114** may be repositioned, for example, in a (fourth) partially open or vent state, as illustrated in FIG. **11**. That is, in at least one example embodiment, the middle pane **114** is movable between a closed position for the middle pane **114** (see FIGS. **1**, **2**, **10A**, and **10B**) and three open positions (see FIGS. **5A** and **5B** illustrate a first open position for the middle pane **114**, FIGS. **6** and **11** illustrates a second open position for the middle pane **114**, FIG. **8** illustrates a third open position for the middle pane **114**).

[0076] In at least one example embodiment, the middle pane **114** may be placed in the overhead position with the top pane, as illustrated in FIGS. **5A** and **5B**, from the closed position for the middle pane **114** by retracting the respective tongue members of the latch assemblies **162A**, **162B** and applying an upward force such that the middle pane moves (i.e., slides) over the top pane **116** and then releasing the a top portion of the second track **140** by retracting the tongue members of the latch assemblies **160A**, **160B** to allow the top portion of the second track **140** with the top pane **112** and the middle pane **114** to move (e.g., swings or pivots) into the overhead position.

[0077] In at least one example embodiment, the middle pane **114** may be slid within the second track **140** between the closed position for the middle pane **114** and the (first) partially open or vent state, as illustrated in FIG. **6**, by retracting the respective tongue members of the latch assemblies **162A**, **162B** and applying a downward force such that the middle pane **114** moves (i.e., slides) behind the bottom pane **116**. The respective tongue members of the latch assemblies **162A**, **162B** may be released and extended to interface with one or more notches **146** to secure the middle pane **114** in the (first) partially open or vent state.

[0078] In at least one example embodiment, the middle pane **114** may be slid within the second track **140** between the closed position for the middle pane **114** and the (second) partially open or vent state, as illustrated in FIG. **8**, by retracting the respective tongue members of the latch assemblies **162A**, **162B** and applying an upward force such that the middle pane moves (i.e., slides) over the top pane **116**. The respective tongue members of the latch assemblies **162A**, **162B** may be released and extended to interface with one or more notches **146** to secure the middle pane **114** in the (second) partially open or vent state.

[0079] As illustrated, for example, in FIGS. **4** and **4A-4C** (and also FIGS. **7** and **9** as discussed below), the first track **130** may include a supporting portion **132** that supports (or receives at least a portion) of the top pane **112**. The first track **130** may further include a notch structure **134** configured (e.g., shaped) to receive a tongue member of respective latch assemblies **160A**, **160B**. When the respective tongue member is (or tongue members are) retracted, for example, using a trigger assembly of the respective latch assembly, the corresponding pane (i.e., the top pane **112**) may be released and made movable. In at least one example embodiment, the top pane **112** may be repositioned, for example, in an overhead position, as illustrated in FIGS. **5A**, **5B**, **10A**, **10B**, and **11**. That is, in at least one example embodiment, the top pane **112** is movable between a closed position for the top pane **112** (see FIGS. **1**, **2**, **6**, and **8**) and an open position for the top pane **112** (see FIGS. **5A**, **5B**, **10A**, **10B**, and **11**). The top pane **112** may be secured in the overhead position by one or more struts **122** and/or by releasing the tongue member of respective latch assemblies **160A**, **160B** to be received within corresponding notches (or like cavities) (not shown). In at least one example embodiment, as mentioned above, the first track **130** and/or a top portion **140B** of the second track **140** may also move into the overhead position to secure the top pane **112** and/or middle pane **114** in the overhead position.

[0080] In at least one example embodiment, the top pane **112** may be placed in the overhead position, as illustrated in FIGS. **5A**, **5B**, **10A**, **10B**, and **11**, from the closed position for the top pane **112** by retracting the tongue members of the latch assemblies **160A**, **160B** to allow the top portion of the second track **140** with the top pane **112** to move (e.g., swings or pivots) into the overhead position.

[0081] FIG. **5A** is a view of the window assembly **110** in a fully open state in accordance with at least one example embodiment of the present disclosure. FIG. **5B** is a perspective view of the window assembly **110** in the fully open state in accordance with at least one example embodiment of the present disclosure. As illustrated, in the fully open state, the middle pane **114** and the top pane **114** are together disposed in the overhead position. In at least one example embodiment, a method for placing the window assembly **110** in the fully open state may include first placing (or securing) the middle pane **114** behind top pane **114** (for example, as illustrated in FIG. **8**) and then placing (or securing) the middle pane **114** and the top pane **112** in the overhead position. In at least one example embodiment, the frame **120** may include one or more struts **122** (e.g., gas struts) that assists with and secure the overhead positioning.

[0082] FIG. **6** is a view of the window assembly **110** in a (first) partially open state in accordance with at least one example embodiment of the present disclosure. As illustrated, in the (first) partially open or vent state, at least part of the middle pane **114** may be stored over (in front of or behind) at least part of the bottom pane **116**.

[0083] FIG. **7** is a cross-sectional view of the window assembly **110** where the window assembly **110** is in the (first) partially open or vent state and at least part of the middle pane **114** is stored over (in front of or behind) at least part of the bottom pane **116**. As illustrated, the middle pane **114** is supported by a supporting portion **142** and a tongue **314** of the respective latch assembly **162A** is received by a notch **146** in the (second) locking portion **144** to secure the middle pane **114** in the (first) partially open or vent state.

[0084] FIG. **8** is a view of the window assembly **110** in a (second) partially open state in accordance with at least one example embodiment of the present disclosure. As illustrated, in the

(second) partially open or vent state, at least part of the middle pane **114** may be stored over (in front of or behind) at least part of the top pane **112**.

[0085] FIG. **9** is a cross-sectional view of the window assembly **110** is in the (second) partially open or vent state and at least part of the middle pane **114** is stored over (in front of or behind) at least part of the top pane **112**. As illustrated, the middle pane **114** is supported by a supporting portion **142** and a tongue of the respective latch assembly **162A** is received by a notch **146** in the (second) locking portion **144** to secure the middle pane **114** in the (second) partially open or vent state.

[0086] FIG. **10A** is a view of the window assembly **110** in a (third) partially open state in accordance with at least one example embodiment of the present disclosure; and FIG. **10B** is a perspective view of the window assembly **110** in a (third) partially open state in accordance with at least one example embodiment of the present disclosure. As illustrated, in the (third) partially open or vent state, at least part of the top pane **112** is stored overhead.

[0087] FIG. **11** is a view of the window assembly **110** in a (fourth) partially open state in accordance with at least one example embodiment of the present disclosure. As illustrated, in the (fourth) partially open or vent state, at least part of the middle pane **114** may be stored over (in front of or behind) at least part of the bottom pane **116** and at least part of the top pane **112** is stored overhead.

[0088] While some example embodiments have been disclosed herein, it should be understood that other variations may be possible. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure, and all such modifications as would be obvious to one skilled in the art are intended to be included within the scope of the following claims.

[0089] Although described with reference to specific examples and drawings, modifications, additions, and substitutions of example embodiments may be variously made according to the description by those of ordinary skill in the art. For example, the described techniques may be performed in an order different with that of the methods described, and/or elements such as the described system, architecture, devices, circuit, and the like, may be connected or combined to be different from the above-described methods, or results may be appropriately achieved by other elements or equivalents.

## Claims

1. A window assembly having two or more open configurations and comprising: a fixed pane; a movable pane; and a third pane, the movable pane being at least partially movable over the fixed pane and at least partially movable over the third pane.
2. The window assembly of claim 1, wherein the window assembly further includes: one or more latch assemblies coupled to the movable pane.
3. The window assembly of claim 2, wherein each of the one or more latch assemblies includes a tongue member that is movable between an extended position and a retracted position.
4. The window assembly of claim 3, wherein the window assembly further includes: a track assembly configured to support the movable pane, the track assembly including a supporting portion and a locking portion, the supporting portion configured to receive at least a portion of the movable pane, and the locking portion including two or more notches configured to receive the tongue member of a respective latch assembly of the one or more latch assemblies.
5. The window assembly of claim 4, wherein the track assembly is a first track assembly, and the window assembly further includes: a second track assembly configured to receive at least a portion of the fixed pane.
6. The window assembly of claim 1, wherein the movable pane is a first movable pane, and the third pane is a second movable pane, the second movable pane movable between a first position and an overhead position.

7. The window assembly of claim 6, wherein the first movable pane is moveable over the second movable pane and into the overhead position with the second movable pane.
8. The window assembly of claim 6, wherein the window assembly further includes: one or more latch assemblies in association with the second movable pane.
9. The window assembly of claim 8, wherein each of the one or more latch assemblies includes a tongue member that is movable between an extended position and a retracted position.
10. The window assembly of claim 9, wherein the window assembly further includes: a track assembly configured to support the second movable pane, the track assembly including a supporting portion and a locking portion, the supporting portion configured to receive at least a portion of the second movable pane, and the locking portion including one or more notches configured to receive the respective tongue member of a respective latch assembly of the one or more latch assemblies.
11. A window assembly having two or more open configurations and comprising: a fixed pane; a first movable pane movable between a closed position and an open position; and a second movable pane movable between a closed position and a first partially open position where at least a portion of the second movable pane is disposed over the fixed pane, between the closed position and a second partially open position where at least a portion of the second movable pane is disposed over the first movable pane, and between the closed position and the open position.
12. The window assembly of claim 11, wherein the window assembly further includes: one or more latch assemblies associated with the first movable pane.
13. The window assembly of claim 12, wherein the each of the one or more latch assemblies includes a tongue member that is movable between an extended position and a retracted position.
14. The window assembly of claim 13, wherein the window assembly further includes: a track assembly configured to support the first movable pane, the track assembly configured to move between the closed position of the first movable pane to the open position of the first movable pane.
15. The window assembly of claim 11, wherein the window assembly further includes: one or more latch assemblies coupled to the second movable pane.
16. The window assembly of claim 15, wherein the each of the one or more latch assemblies includes a tongue member that is movable between an extended position and a retracted position.
17. The window assembly of claim 16, wherein the window assembly further includes: a track assembly configured to support the second movable pane, the track assembly including a supporting portion and a locking portion, the supporting portion configured to receive at least a portion of the second movable pane, and the locking portion including one or more first notches corresponding with the closed position, one or more second notches corresponding with the first partially open position, one or more third notches corresponding with the second partially open position, and one or more fourth notches corresponding with the open position, each notch of the one or more first notches, of the one or more second notches, of the one or more third notches, and of the one or more fourth notches is configured to receive the respective tongue member of a respective latch assembly of the one or more latch assemblies.
18. A window assembly having two or more open configurations and comprising: a fixed pane; a first pane assembly including a first movable pane that is movable between a closed position and an overhead position, a first latch assembly associated with the first movable pane, and a first track assembly configured itself to move between the closed position of the first movable pane to the overhead position of the first movable pane; and a second pane assembly including a second movable pane that is movable between a closed position and a first partially open position where at least a part of the second movable pane is disposed over a portion of the fixed pane, between the closed position and a second partially open position where at least a part of the second movable pane is disposed over a portion of the first movable pane, and between the closed position and the overhead position, a second latch assembly coupled to the second movable pane, and a second

track assembly including a supporting portion configured to receive at least a portion of the second movable pane, and a locking portion including four or more notches configured to receive a portion of the second latch assembly.

**19.** The window assembly of claim 18, wherein a strut enables movement of the first track assembly between the closed position of the first movable pane and the overhead position of the first movable pane.

**20.** The window assembly of claim 18, wherein the second latch assembly includes a tongue member that is movable between an extended position and a retracted position, and the four or more notches are configured to receive at least a portion of the tongue member.

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